

ANDROID APPLICATION DEVELOPMENT LAB

MCA II A (2)

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25/SCA/MCAN/002

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5	Create an application that uses a text file to store user names and passwords (tab separated fields and one record per line) When the user submits a login name and password through a screen, the details should be verified with the text file data and if they match, show a dialog saying that login is successful. Otherwise, show the dialog with Login Failed message.	
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Program 1: Create an application that takes the name from a text box and shows hello message along with the name entered in text box, when the user clicks the OK button.

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent" android:orientation="vertical"
    android:padding="16dp">

    <EditText android:id="@+id/et1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter text here"
        android:inputType="text" />

    <Button android:id="@+id/btn1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Click Me"
        android:layout_marginTop="16dp" />

    <TextView android:id="@+id/tv1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello Android!"
        android:textSize="16sp"
        android:layout_marginTop="16dp" />

</LinearLayout>
```

MainActivity.java:

```
package com.example.enter_username;
```

```

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button btn;
    EditText et;
    TextView tv;

    @Override protected void onCreate(Bundle
savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btn = findViewById(R.id.btn1); et =
            findViewById(R.id.et1); tv =
            findViewById(R.id.tv1);

        btn.setOnClickListener(new View.OnClickListener() {
            @Override public void
            onClick(View v) {

                String text = et.getText().toString(); String

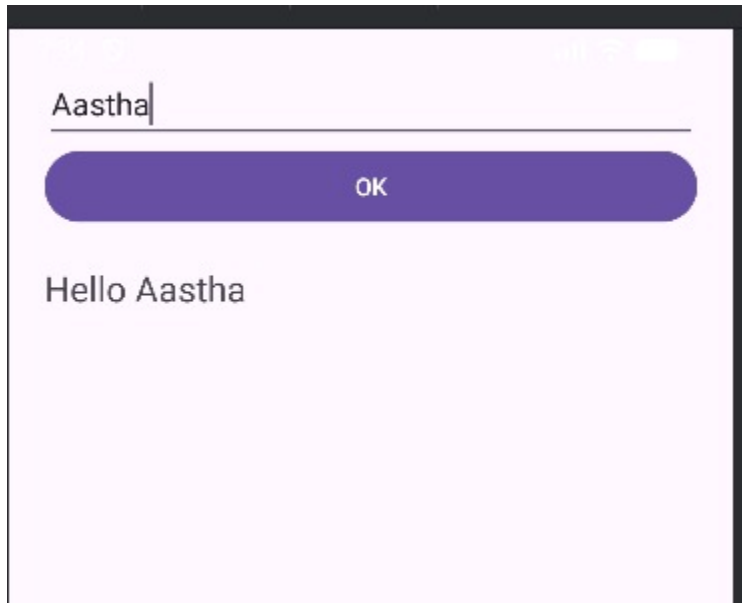
                message = "Hello " + text;

                tv.setText(message);

                Toast.makeText(MainActivity.this, message, Toast.LENGTH_SHORT).show();
            }
        });
    }
}

```

Output:



Program 2: Create a screen that has input boxes for User Name, Password, Address, Gender (radio buttons for male and female), Age (numeric), Date of Birth (Date Picket), State (Spinner) and a Submit button. On clicking the submit button, print all the data below the Submit Button. Use 1. Linear Layout 2. Relative Layout and 3. Grid Layout or Table Layout.

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?><ScrollView
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent" android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">

        <!-- TABLE LAYOUT -->
        <TableLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content">

            <!-- Username -->
            <TableRow>
                <TextView android:text="Username:"/>
                <EditText android:id="@+id/etName"
                    android:layout_width="match_parent"
                    android:inputType="text"/>
            </TableRow>

            <!-- Password -->
            <TableRow>
                <TextView android:text="Password:"/>
                <EditText android:id="@+id/etPassword"
                    android:layout_width="match_parent"
                    android:inputType="textPassword"/>
            </TableRow>

        </TableLayout>

    </LinearLayout>

</ScrollView>
```

```
<!-- Address -->
<TableRow>
    <TextView android:text="Address:"/>
    <EditText android:id="@+id/etAddress"
        android:layout_width="match_parent"
        android:inputType="textMultiLine"/>
</TableRow>

<!-- Gender -->
<TableRow>
    <TextView android:text="Gender:"/>
    <RadioGroup android:id="@+id/radioGroup">

        <RadioButton android:id="@+id/rbMale"
            android:text="Male"/>

        <RadioButton android:id="@+id/rbFemale"
            android:text="Female"/>
    </RadioGroup>
</TableRow>

<!-- Age -->
<TableRow>
    <TextView android:text="Age:"/>
    <EditText android:id="@+id/etAge"
        android:layout_width="match_parent"
        android:inputType="number"/>
</TableRow>

<!-- DOB -->
<TableRow> de<TextView android:text="DOB:"/>
    <DatePicker android:id="@+id/datePicker"
        android:layout_width="wrap_content"/>
</TableRow>

<!-- State -->
<TableRow>
    <TextView android:text="State:"/>
    <Spinner android:id="@+id/spinnerState"/>
```

```

        </TableRow>

    </TableLayout>

    <!-- SUBMIT BUTTON -->
    <Button android:id="@+id/btnSubmit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Submit"
        android:layout_marginTop="20dp"/>

    <!-- OUTPUT -->
    <TextView android:id="@+id/tvResult"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:textSize="16sp"/>

</LinearLayout>

</ScrollView>

```

MainActivity.java:

```

package com.example.formapp;

import
android.os.Bundle; import
android.view.View; import
android.widget.*;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extend AppCompatActivity {
    EditText etName, etPassword, etAddress, etAge;
    RadioGroup radioGroup; Spinner
    spinner;
    DatePicker datePicker;
    Button btnSubmit;
    TextView tvResult;

    String[] states = {"Haryana", "Delhi", "Punjab", "UP", "Rajasthan"};
    @Override

```



```

protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    etName      = findViewById(R.id.etName);
    etPassword  = findViewById(R.id.etPassword);
    etAddress   = findViewById(R.id.etAddress);
    etAge       = findViewById(R.id.etAge);
    radioGroup  = findViewById(R.id.radioGroup);
    spinner     = findViewById(R.id.spinnerState);
    datePicker  = findViewById(R.id.datePicker);
    btnSubmit   = findViewById(R.id.btnSubmit);
    tvResult    = findViewById(R.id.tvResult);

    // Spinner setup
    ArrayAdapter<String> adapter = new ArrayAdapter<>(
        this,
        android.R.layout.simple_spinner_item,
        states
    );
    adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);
    spinner.setAdapter(adapter);

    btnSubmit.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {

            String name = etName.getText().toString();
            String pass = etPassword.getText().toString();
            String address = etAddress.getText().toString(); String
            age = etAge.getText().toString();

            int selectedId = radioGroup.getCheckedRadioButtonId();
            RadioButton rb = findViewById(selectedId);
            String gender = (rb != null) ? rb.getText().toString() : "Not Selected";

            int day = datePicker.getDayOfMonth();
            int month = datePicker.getMonth() + 1;
            int year = datePicker.getYear();

            String dob = day + "/" + month + "/" + year;

            String state = spinner.getSelectedItem().toString();

            String result =
                "Name: " + name + "\n" +
                "Password: " + pass + "\n" +

```

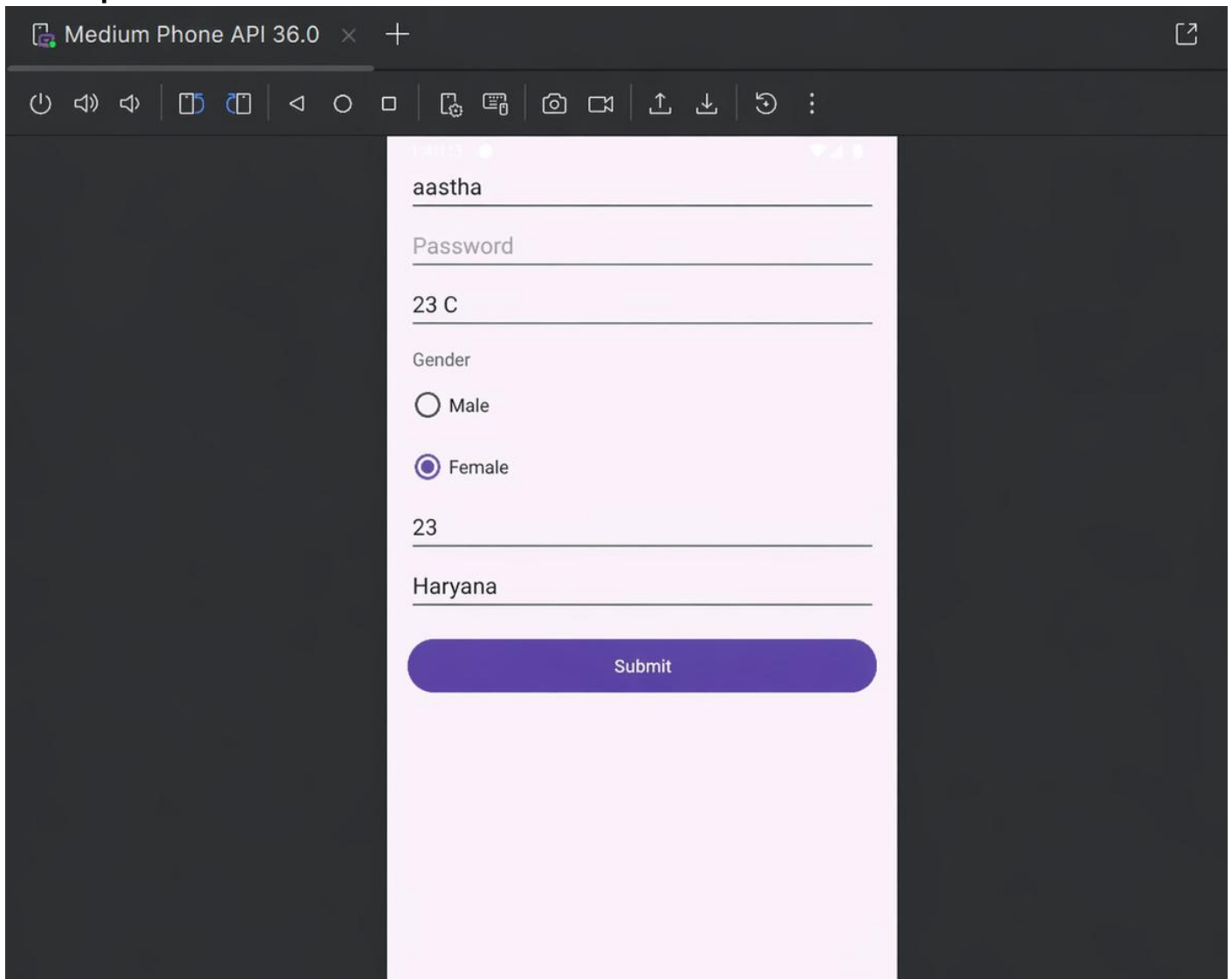
```

        "Address: " + address + "\n" +
        "Gender: " + gender + "\n" +
        "Age: " + age + "\n" +
        "DOB: " + dob + "\n" + "State:
        " + state;

        tvResult.setText(result);
    }
});
}
}

```

Output:



Program 3: Develop an application that inserts some notifications into Notification area and whenever a notification is inserted, it should show a toast with details of the notification.

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent" android:gravity="center"
    android:orientation="vertical" android:padding="20dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Notification App"
        android:textSize="22sp"
        android:textStyle="bold"
        android:layout_marginBottom="30dp"/>

    <Button android:id="@+id/btnNotify"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Show Notification"/>

</LinearLayout>
```

MainActivity.java:

```
package com.example.notificationapp;
```

```

import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.os.Build;      import
android.os.Bundle;          import
android.widget.Button;       import
android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity; import
import androidx.core.app.NotificationCompat;

public class MainActivity extends AppCompatActivity {

    Button btnNotify;
    String CHANNEL_ID = "my_channel";

    @Override protected void onCreate(Bundle
savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnNotify = findViewById(R.id.btnNotify);

        btnNotify.setOnClickListener(v -> {

            String title = "Hello User";
            String message = "This is your notification";

            // Toast showing notification details
            Toast.makeText(this,
                "Notification: " + title + " - " + message,
                Toast.LENGTH_LONG).show();

            NotificationManager manager =
                (NotificationManager) getSystemService(NOTIFICATION_SERVICE);

            // For Android 8+
            if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
                NotificationChannel channel = new
                    NotificationChannel(
                        CHANNEL_ID,
                        "My Channel",
                        NotificationManager.IMPORTANCE_DEFAULT);

                manager.createNotificationChannel(channel);
            }
        })
    }
}

```

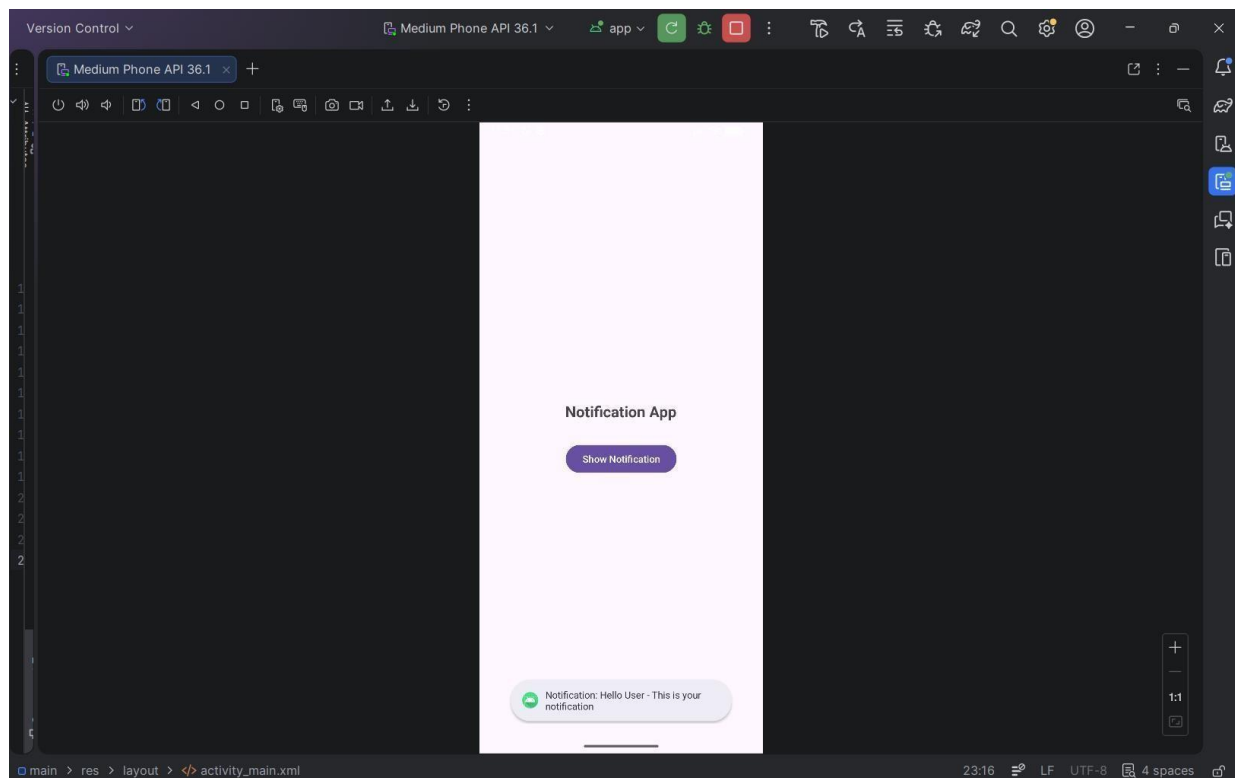
```

        NotificationCompat.Builder builder = new
            NotificationCompat.Builder(this, CHANNEL_ID)
                .setContentTitle(title)
                .setContentText(message)
                .setSmallIcon(android.R.drawable.ic_dialog_info)
                .setAutoCancel(true);

        manager.notify(1, builder.build()); });
    }
}

```

Output:



Program 4: Develop an application that uses a menu with 3 options for dialing a number, opening a website and to send an SMS. On selecting an option, the appropriate action should be invoked using intents.

activity_main.xml

:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent" android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Menu Intent App"
        android:textSize="22sp"
        android:textStyle="bold"/>

</LinearLayout>

```

menu_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">

    <item
        android:id="@+id/menu_call"
        android:title="Dial
        Number"/>

    <item android:id="@+id/menu_web"
        android:title="Open
        Website"/>

    <item
        android:id="@+id/menu_sms"
        android:title="Send SMS"/>

</menu>

```

MainActivity.java:

```

package com.example.intentmenu;

import android.content.Intent;
import android.net.Uri; import
android.os.Bundle;

```

```
import android.view.Menu; import android.view.MenuItem;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState); setContentView(R.layout.activity_main);
    }

    // Create menu @Override public boolean onCreateOptionsMenu(Menu menu) {
    getMenuInflater().inflate(R.menu.menu_main, menu); return true;
    }

    // Handle menu clicks
    @Override public boolean onOptionsItemSelected(MenuItem item) {

        int id = item.getItemId();

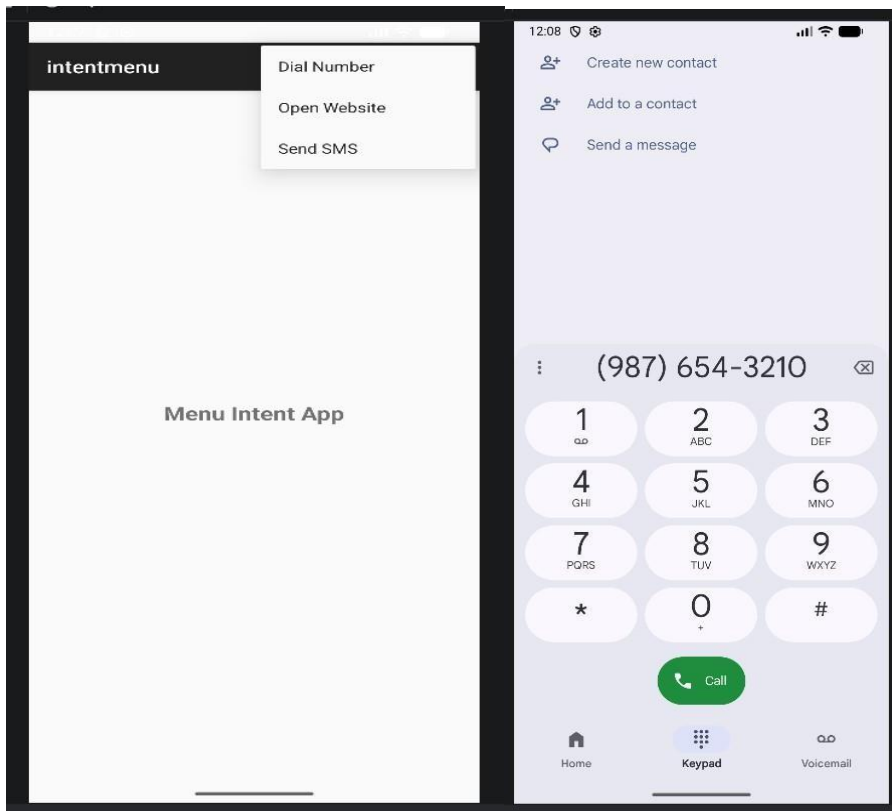
        // Dial Number
        if (id == R.id.menu_call) {
            Intent intent = new Intent(Intent.ACTION_DIAL);
            intent.setData(Uri.parse("tel:9876543210")); startActivity(intent);
            return true;
        }

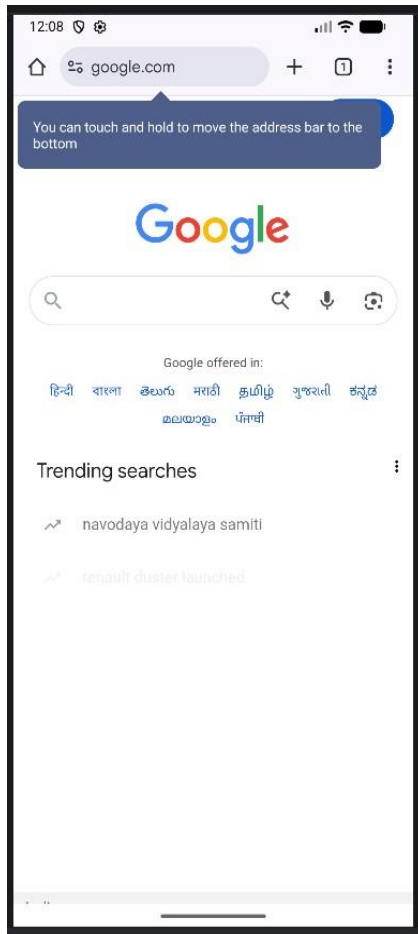
        // Open Website
        if (id == R.id.menu_web) {
            Intent intent = new Intent(Intent.ACTION_VIEW);
            intent.setData(Uri.parse("https://www.google.com")); startActivity(intent);
            return true;
        }

        // Send SMS
        if (id == R.id.menu_sms) {
            Intent intent = new Intent(Intent.ACTION_VIEW);
            intent.setData(Uri.parse("sms:9876543210")); intent.putExtra("sms_body",
            "Hello from my app!");
        }
    }
}
```

```
        startActivity(intent);  
        return true;  
    }  
    return super.onOptionsItemSelected(item); }  
}
```

Output:





Program 5: Create an application that uses a text file to store user names and passwords (tab separated fields and one record per line) When the user submits a login name and password through a screen, the details should be verified with the text file data and if they match, show a dialog saying that login is successful. Otherwise, show the dialog with Login Failed message.

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:gravity="center" android:orientation="vertical" android:padding="20dp">
```

```

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Login System"
    android:textSize="22sp"
    android:textStyle="bold"
    android:layout_marginBottom="30dp"/>

<EditText android:id="@+id/etUser"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Username"/>

<EditText android:id="@+id/etPass"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Password"
    android:inputType="textPassword"
    android:layout_marginTop="10dp"/>

<Button android:id="@+id/btnLogin"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Login"
    android:layout_marginTop="20dp"/>

</LinearLayout>

```

MainActivity.java:

```

package com.example.filelogin;

import
android.app.AlertDialog; import
android.os.Bundle;      import
android.widget.Button;   import
android.widget.EditText;

import androidx.appcompat.app.AppCompatActivity;

import java.io.BufferedReader; import
java.io.InputStreamReader;

public class MainActivity extends AppCompatActivity
{

```

```

EditText etUser, etPass;
Button btnLogin;

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState); setContentView(R.layout.activity_main);

    etUser = findViewById(R.id.etUser); etPass
    = findViewById(R.id.etPass); btnLogin =
    findViewById(R.id.btnLogin);

    btnLogin.setOnClickListener(v -> {

        String inputUser = etUser.getText().toString(); String
        inputPass = etPass.getText().toString();

        boolean isValid = false;

        try {
            BufferedReader reader = new BufferedReader( new
                InputStreamReader(getAssets().open("users.txt"))
            );

            String line;

            while ((line = reader.readLine()) != null) { String[]

                parts = line.split("\t");

                if (parts.length == 2) {
                    String fileUser = parts[0]; String
                    filePass = parts[1];

                    if (inputUser.equals(fileUser) && inputPass.equals(filePass)) {
                        isValid = true; break;
                    }
                }
            }

            reader.close();

        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}

```

```
if (isValid) { showDialog("Login Successful");
                } else { showDialog("Login Failed"); }

    });
}

private void showDialog(String message) { new AlertDialog.Builder(this)
    .setTitle("Result")
    .setMessage(message)
    .setPositiveButton("OK", null)
    .show();
} }
```

Users.txt:

```
priya 123
sonam 456
aashu 789
kasak 1011
aastha 1022
```

Output:

Login System

aastha

....

Login



Login System

Result

Login Successful

OK

Login



Program 6: Create an activity to navigate to different activities through Intents- Implicit and Explicit

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent" android:gravity="center"
    android:orientation="vertical" android:padding="20dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Intent Demo"
        android:textSize="22sp"
        android:textStyle="bold"
        android:layout_marginBottom="30dp"/>

    <Button android:id="@+id/btnExplicit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Go to Second Activity"
        android:layout_marginBottom="15dp"/>

    <Button android:id="@+id/btnWebsite"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Open Website"
        android:layout_marginBottom="15dp"/>

    <Button android:id="@+id/btnDial"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Dial Number"/>

</LinearLayout>
```

MainActivity.java:

```

package com.example.intentdemo;

import com.example.intentdemo.SecondActivity;
import android.content.Intent;    import
android.net.Uri;    import android.os.Bundle;
import android.widget.Button;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button btnExplicit, btnWebsite, btnDial;

    @Override protected void onCreate(Bundle
savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnExplicit = findViewById(R.id.btnExplicit);
        btnWebsite = findViewById(R.id.btnWebsite);
        btnDial = findViewById(R.id.btnDial);

        // ◇ EXPLICIT INTENT
        btnExplicit.setOnClickListener(v -> {
            Intent intent = new Intent(MainActivity.this, SecondActivity.class);
            startActivity(intent);
        });

        // ◇ IMPLICIT INTENT - OPEN WEBSITE btnWebsite.setOnClickListener(v
-> {
            Intent intent = new Intent(Intent.ACTION_VIEW);
            intent.setData(Uri.parse("https://www.google.com"));
            startActivity(intent);
        });

        // ◇ IMPLICIT INTENT - DIAL btnDial.setOnClickListener(v
-> {
            Intent intent = new Intent(Intent.ACTION_DIAL);
            intent.setData(Uri.parse("tel:9876543210"));
            startActivity(intent);
        });
    }
}

```

activity_second.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent" android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="This is Second Activity"
        android:textSize="20sp"
        android:textStyle="bold"/>

</LinearLayout>
```

SecondActivity.java:

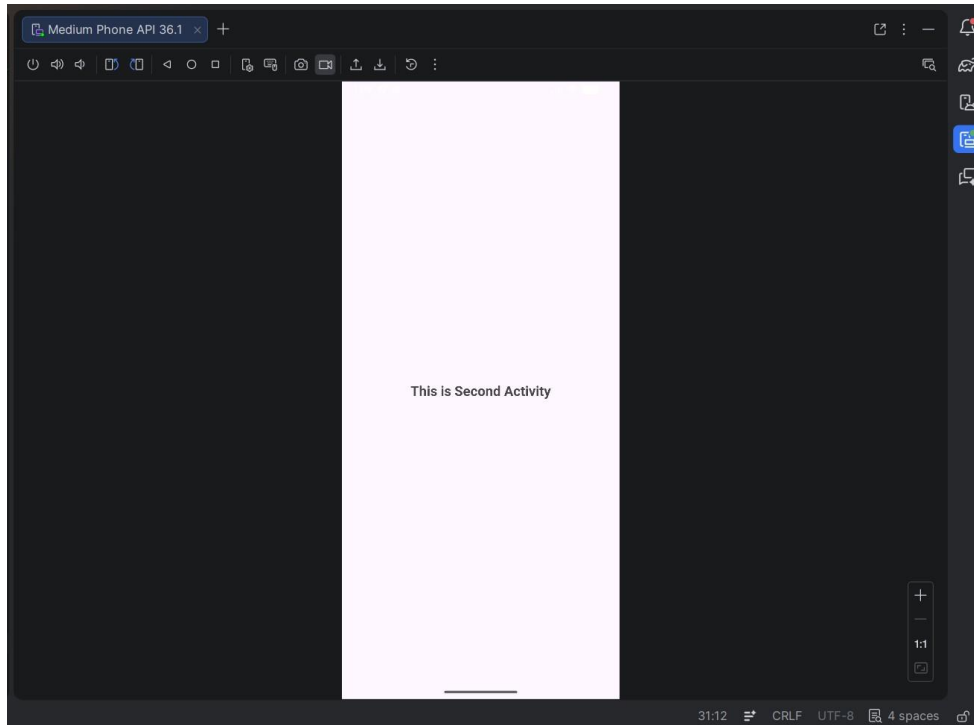
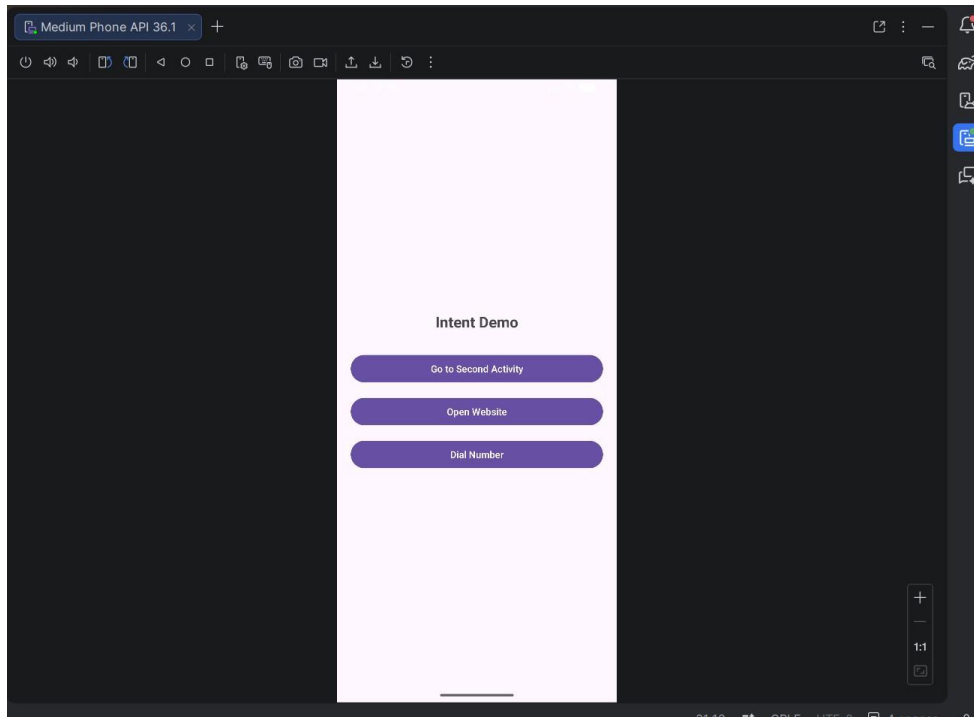
```
package com.example.intentdemo;

import android.os.Bundle; import
androidx.appcompat.app.AppCompatActivity;

public class SecondActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState)
    { super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_second); } }
```

Output:



Program 7: Create a activity to create option Menu.

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent" tools:context=".MainActivity">

    <TextView            android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Option            Menu"
        android:textSize="22sp"
        android:textStyle="bold"
        android:layout_centerInParent="true" />

</RelativeLayout>

```

MainActivity.java:

```

package com.example.optionmenu;

import android.os.Bundle; import android.view.Menu;
import android.view.MenuItem; import
android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState); setContentView(R.layout.activity_main); }

    // Step 1: Create Option Menu
    @Override public boolean onCreateOptionsMenu(Menu menu) { menu.add(0, 1, 0,
        "Settings"); menu.add(0, 2, 0, "Profile");

```

```
        menu.add(0, 3, 0, "Logout"); return
        true;
    }

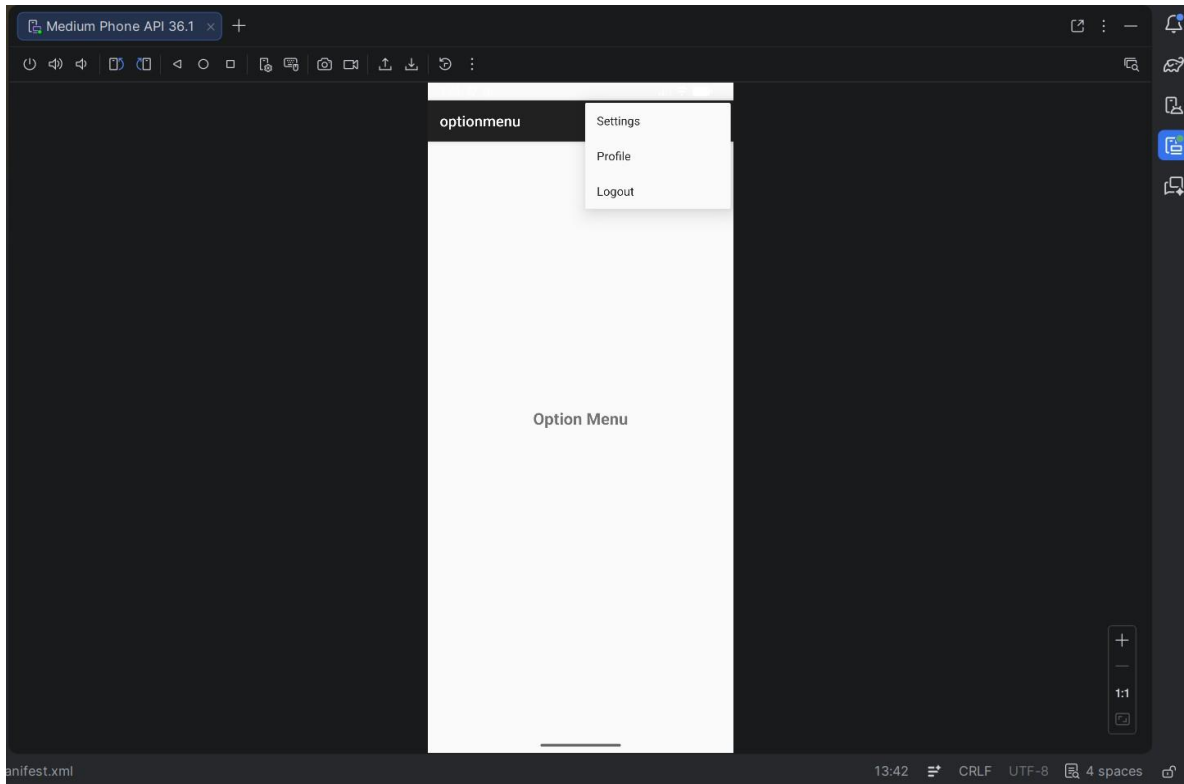
    // Step 2: Handle Menu Clicks
    @Override public boolean
    onOptionsItemSelected(MenuItem item) {
switch (item.getItemId()) { case 1:
    Toast.makeText(this, "Settings Clicked", Toast.LENGTH_SHORT).show();
    return true;

case 2:
    Toast.makeText(this, "Profile Clicked", Toast.LENGTH_SHORT).show();
    return true;

case 3:
    Toast.makeText(this, "Logout Clicked", Toast.LENGTH_SHORT).show();
    return true;

default:
    return super.onOptionsItemSelected(item);
        }
    }
}
```

Output:



Program 8: Create an activity using Button Controls- CheckBox, RadioButtons, Radio Groups

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent" android:orientation="vertical"
    android:padding="20dp">

    <!-- CheckBox -->
    <CheckBox
```

```

        android:id="@+id/checkbox"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content" android:text="Accept
        Terms and Conditions" />

<!-- RadioGroup -->
<RadioGroup android:id="@+id/radioGroup"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="20dp">

    <RadioButton android:id="@+id/radioMale"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Male" />

    <RadioButton
        android:id="@+id/radioFemale"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Female" />

</RadioGroup>

<!-- Button -->
<Button android:id="@+id/btnSubmit"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Submit"
    android:layout_marginTop="20dp"/>

</LinearLayout>

```

MainActivity.java:

```

package com.example.buttoncontrols;

import android.os.Bundle; import android.view.View; import
android.widget.Button; import android.widget.CheckBox; import
android.widget.RadioButton; import android.widget.RadioGroup;
import android.widget.Toast;

```

```

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    CheckBox checkBox;
    RadioGroup radioGroup;
    Button btnSubmit;

    @Override protected void onCreate(Bundle
savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        checkBox = findViewById(R.id.checkBox);
        radioGroup = findViewById(R.id.radioGroup);
        btnSubmit = findViewById(R.id.btnSubmit);

        btnSubmit.setOnClickListener(new View.OnClickListener() {
            @Override public void
onClick(View v) {

                // CheckBox state boolean isChecked =
checkBox.isChecked();

                // RadioButton selected int selectedId =
radioGroup.getCheckedRadioButtonId();

                if (selectedId == -1) {
                    Toast.makeText(MainActivity.this, "Select Gender",
Toast.LENGTH_SHORT).show();
                    return;
                }

                RadioButton selectedRadio = findViewById(selectedId);
                String gender = selectedRadio.getText().toString();

                // Show result
                String message = "Accepted: " + isChecked + "\nGender: " + gender;

                Toast.makeText(MainActivity.this, message, Toast.LENGTH_LONG).show();
            }
        });
    }
}

```

Output:

☒ Accept Terms and Conditions

☐ Male

☒ Female

Submit